

RENANCE

PETROLEUM TRAINING INSTITUTE, EFFURUN

CET 123 — Past Questions & Materials

by Renance

Course: CET 123

Institution: Petroleum Training Institute, Effurun

Papers in this pack: 1

Levels: ND1

This study pack was compiled and restructured by Renance for CET 123 students at Petroleum Training Institute, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 1 — CET 123 SELABUS

Level ND1 • First Semester • Handout

PROGRAMME: NATIONAL DIPLOMA IN CODE: CTE 123 CONTACT HOUR: 45 HRS

COMPUTER

ENGINEERING TECHNOLOGY

COURSE: COMPUTER/ELECTRONIC MAINTENANCE Semester:2 Theoretical:1 hour/week
AND REPAIRS

Year: 1 Pre-requisite: Practical: 2 hrs

Goal: This course is designed to provide students with practical knowledge and skills in maintenance and repairs of

electronic/computer equipment.

GENERAL OBJECTIVES:

On completion of this module, the student should be able to:

1. Understand the general use of tools and testing instruments.
2. Understand cabling, jointing soldering and de-soldering techniques.
3. Know different electronic circuit components
4. Use manufactures service manual and circuit wiring diagrams.
5. Maintain GSM phones.
6. Use of Uninterruptible Power Supply (UPS) and Automatic Voltage Regulators (AVR)

Theoretical Content Practical Content

General Objective 1: Understand the general use of tools and testing instruments

Week Specific Learning Teacher's Learning Resources Specific Teacher's Evaluation

Outcomes Activities Learning Activities

Outcomes

1-2 1.1 Identify □ List, sketch □ Textbooks □ Carry out □ Teacher Explain the use
the and state □ Instrum identificatio demonstrates of Multi-tester;
following the ent n of items the use of Transistor tester
test applications catalogs in 1.1 items in 1.1 Oscilloscope and
instrument of each test □ Instrument □ Show □ Ask students Electronic
s: instrument in user competenc to identify voltmeter
i. Multi-tester; 1.1 manuals. e in the use and instruments
ii. Transist □ Describe with □ Multi-tester; of items in demonstrate
or the students □ Transistor 1.1 the use of
tester; in more tester; items in 1.1
iii. Oscilloscope details the □ Oscilloscope;
; use items in □ Electronic
iv. Electronic 1.1. voltmeter
voltmeter instrumen
instrumen ts.
ts. □ Practic

1.2 Explain the al
uses of the manua
items in ls
1.1 above

General Objective 2: Understand cabling jointing soldering and de-soldering techniques

Week Specific Learning Teacher's Learning Resources Specific Teacher's Evaluation

Outcomes Activities Learning Activities

Outcomes

3-6 2.1 Explain the □ Describe □ Textbooks □ Perform □ Demonstrate 1 Explain the types of
types of cables, □ Journals the the cables used in;

cables used discuss in □ Soldering iron procedu procedure -Power supply

in; details the □ Lead re and and Communication

i) Power what is meant □ Lead sucker techniqu techniques in between systems.

supply ii) by the term □ Cutter es in 2.3 2.2 and 2.3. Communication

Communicati cabling as □ Vero boards and 2.3. □ Ask students between systems

on between well as steps □ components □ Wire up to and peripherals.

systems. for cabling required for and demonstrate □

iii) □ Describe with selected solder the

Communicati illustration circuits compon procedure

on between details of □ RJ45connector en t to and

systems and items in 2.2, make techniques

peripherals. 2.3 and 2.4 simple in

2.2 and 2.3

□ Guide

students to

carry out

the

2.2 Outline □ Crimping electronic procedure

cabling □ tools circuits. and

procedure □ Communicati □ Carry techniques

and on cables out de- in 2.4

practice. □ Practic solderi

2.3 Explain the al ng

types of manua □ Carry

cables, choice ls out each

and methods techniqu

of testing, as es in 2.4

well as the

instruments

used for

testing:

i) Twisted

pair

cables

ii) Coaxial

cables RS-

232

standard

communicat

io n cables

2.4 Explain

n the

following:

a. Jointin

g

techniqu

es;

b. Soldering

and

disordering

techniques;

c. Crimping

and

fastening

method.

General Objective 3: Know different circuit components

Week Specific Learning Teacher's Learning Resources Specific Teacher's Evaluation

Outcomes Activities Learning Activities

Outcomes

7-8 3.1 Identify □ Explain how □ Textbooks □ Determini □ Ask students □ What is the uses of

values of to determine $\square$ Assorte ne the to determine resistors and

resistors and values of d values the values of capacitors using

capacitors resistors and Resistor of resistors and colour codes

using colour capacitors rs resistors capacitors Explain the value of

codes. using colour □ Assorted and using colour the following

3.2 List different codes. Capacito capacito codes. electronic

types of □ List different rs rs components;

resistors □ Assorted

(carbon, wire- types of Transistors using i. Transistors;

wound, ketal resistors and □ Assorted colour □ Demonstra ii. Diodes;

oxide etc. and capacitors in diodes codes. te and i. Integrated

capacitors. 3.2 □ Assorted □ Carry carry out circuit

3.3 Identify □ Using logic ICs out test test for (IC's);
 the catalog, □ Digital for each each ii. Resistors by
 following explain the multimeter compon component their
 electronic various ers en t in in preferred
 componen available □ Practic 3.3 3.3 using values and
 ts; preferred al using techniques power rating.
 i. Transistors; values of manua techniqu in 3.5 iii. Capacitors by
 ii. Diodes; items in 3.3 Is es in 3.5 their working
 i. Integrated voltage and
 circuit types.
 (IC's);
 ii. Resistors
 by their
 preferred
 values and
 power
 rating.
 iii. Capacitors
 by their
 working
 voltage
 and types.

3.4 Explain open-
 circuit and
 short-circuit
 defects in
 components
 listed in 3.3.

3.5 Outline
 various
 methods of
 testing
 components:
 a. In-circuit.
 b. Out of
 circuit.

General Objective 4: Understand the use of manufactures service manual and circuit wiring diagrams

Week Specific Teacher's Learning Resources Specific Teacher's Evaluation

Learning Activities Learning Activities

Outcomes Outcomes

9-11 4.1 Explain □ List, explain □ Textbooks, □ Carry out □ Demonstrate □ List types of fault observation □ manufactu fault and instruct trouble shooting tracing in test methods res tracing students to and fault isolating circuits listed in 4.1 service, by carry out techniques and

4.2 Outline □ manual observati fault tracing observation test trouble and circuit on by method shooting wiring methods observation and fault diagrams. □ Perform methods

isolating □ manufactur test to □ Demonstrate techniques. es service identify and guide

4.3 List manual faulty students to observation □ circuit compone identify test method: wiring nt s by faulty

- i. Visual; diagrams measure module by
- ii. Touch; □ logic probe me nt measuremen
- iii. Smell; □ oscilloscope (voltage t (voltage
- iv. Hearing. □ Digital and and

4.4 Explain D.C multimet resistanc resistance and ers e test). test).

A.C signal □ Replacea □ Replace testing. ble faulty

4.6 Explain modules compone stage or □ Practic nt module by al s.

substitution. manua ls

General Objective 5: Know the maintenance of GSM phones

Week Specific Teacher's Learning Specific Teacher's Evaluation

Learning Activities Resources Learning Activities

Outcomes Outcomes

12-13 5.1 Identify □ Describe the □ Textbooks, □ Perform □ Demonstrate □ Describe common various function and □ Catalogs the and guide GSM hardware mobile use of hands □ Phone following students to problems related to phone free/headset, manua on GSM solve mouthpiece, (GSM) earpiece, ls phones: common earpiece, charging accessories external □ Good and - Troubl hardware port, keyboard and and their Bluetooth, Scrap mobile es problems damage screen, function chargers, phones hootin stated in 5.2 vibrator.

5.2 Explain the batteries, etc, □ Workstation g, □ Demonstrate □ Describe common various □ List and □ Data cable - Disman and guide GSM software mobile describe □ Phone tli ng, students to problems such as

phone common GSM manua - Assemb resolve SIM rejection,
 (GSM) hardware Is le and common phone lock, invalid
 accessories problems □ Precision set - Test hardware SIM , hanging,
 and their listed in □ Allen key set □ Perform
 functions fault

5.3 Identify finding

common GSM using
 hardware

problems 5.2 □ Magnifyi software. problems
 related to □ List and ng desk stated in 5.3

mouthpiece, describe lamp

earpiece, common GSM □ Service

charging software provider

port, problems listd chart codes

keyboard and in 5.3 □ Ultrason

damage ic

screen, cleaner

vibrator, etc □ Flashing

5.4 Identify and

common GSM unlocking

software devices/co

problems mp uter

such as SIM softwares

rejection, □ Practic

phone lock, al

invalid SIM , manua

hanging, ls

restarting

etc.

General Objective 6: Understand the use of Uninterruptible Power supply (UPS) and Automatic
 voltage

regulators (AVR)

Week Specific Learning Teacher's Learning Specific n ETvaelaucahteior'ns

Outcomes Activities Resources Learni S pe c i f i c

g O u t c o m Activities

Lees arning

Outcomes

14-15 6.1 Briefly □ Explain the □ Textbooks □ Demonstr □ Demonstrate What is the process

explain the applications of □ UPS ate faults the common of cooling and

operation of UPS with □ AVR diagnoses faults in UPS what is the

a UPS and emphasizes on □ Good and in UPS such as bad essence of

AVR for battery usages, bad UPS and AVR battery, having good steady charging, and batteries power cord ventilation and power effect of UPS □ Good and open circuit, cooling systems. supply in over loading open etc

computer □ Describe circuited system. different power

6.2 Explain the methods of cords

process of cooling and □ Practical cooling and ventilation in manuals/gui

the essence computer de

of having power system

good

ventilation

and cooling

systems.

6.3 Explain

maintenance

of batteries

and

battery

chargers

Type of Assessment Purpose and Nature of Assessment Weighting

Examination Final Examination (written) to assess knowledge 40

and understanding

Test At least 1 progress test for feedback. 10

Practical / Projects To be assessed by the teacher 40

Course work/ assignment To be assessed by the teacher 10

Total 100